PROJECT MAP · ARXIV 2605.06527 · AS OF 12 OCT 2026

Does Write-Side State Adjudication Survive Small Models?

Reproducing and stress-testing STALE and CUPMem.

done · to do · open question

STALE / CUPMem reproduction

Reproduction + stress test · MLRC / TMLR · ACL/EMNLP workshops

Dataset 4

- `T1_T2_400_FULL.json`, 306 MB, CC BY 4.0
- 200 Type I + 200 Type II instances, 50 sessions each
- Gold evidence: `[idx_old, idx_new]`, about 27 sessions apart
- Mean 151.7K tokens (max 164.9K); every instance exceeds 128K

Evaluation protocol 4

- One joint judge call per instance, temperature 0, JSON mode
- Judge: Gemini-3.1-flash-lite (exact model id unconfirmed)
- Overall = unweighted mean of 6 (type × dimension) cells
- Truncation to 128K: head ≤2, tail ≤2, then random between evidence sessions

Open questions 6

- **CUPMem session mode**: default `relevant_only` ingests only gold sessions. Which mode gave 68%? $0.37/instance cost points to `full`
- Target-model temperature not stated in the paper
- Judge errors silently scored as FAIL in official code
- KEEP/STALE/REPLACE/UNKNOWN (§5) vs ADD/REFINE/REPLACE/NO_OP (App. F.2)
- Embedding model `all-MiniLM-L6-v2` only in README
- Find and read *The Memory Trust Gap*

Deviations logged 4

- **D1** truncation RNG seeded by (seed, uid, dim)
- **D2** judge errors recorded and re-judged on resume
- **D3** target temperature 0 (official: API default)
- **D4** tiktoken + o200k fallback, same as official

Findings so far 5

- `gpt-4o-mini-2024-07-18` still served, pinned
- 200K TPM limit ≈ 1 B0 call/min: dev 2.5 h, full E1 ≈ 20 h
- Truncation parity: estimate within 2 tokens of billed
- Smoke test: 4/4 instances, 1.49M input tokens, ≈ $0.23
- PR answers accept false premises; SR reaffirms old state

Related work 2

- **StateAuditor** (2608.01619): readout-side audit, +5.0 on STALE
- **TEPA** (2608.07429): revokes keyed precedents; flags multi-hop as out of scope, supports RQ5

Source paper 6

- **STALE**, Chao et al., May 2026: can agents tell when memories are no longer valid?
- **Implicit conflict**: a later observation invalidates an old memory with no explicit negation
- 400 scenarios, 1,200 queries, contexts up to 150K tokens
- Probes: **SR** state resolution · **PR** premise resistance · **IPA** policy adaptation
- Best frontier model 55.2% overall; GPT-4o-mini B0 8.7%
- **CUPMem** prototype: write-time state consolidation + propagation-aware search, 68.0%

Research questions 5

- **RQ1** Reproduction: paper vs official code vs our reimplementation
- **RQ2** Small 3–9B open adjudicators
- **RQ3** Stage-wise failure attribution
- **RQ4** Ablations and cost
- **RQ5** 2-hop cascades

Systems under test 6

- **B0** full context + evidence-preserving truncation
- Official CUPMem wrapper (reference run)
- **M** independent CUPMem reimplementation
- **B1** naive RAG
- **B2** Mem0
- StateAuditor as a possible readout-side baseline

Harness 7

- `data/loader.py` records → Instance/Session/Probe, stratified subsets
- `llm/gateway.py` sqlite cache, JSON decoding, usage, TPM limiter
- `memory/base.py` MemorySystem interface + official prompts
- `eval/judge.py` official joint judge, verbatim prompt
- `eval/stats.py` 6-cell mean, bootstrap CI, McNemar
- `runner.py` config-driven, resumable, parallel, `--no-judge`
- Diagnostics / attribution module

Timeline 5

- 13-week blueprint, started 28 Sep 2026
- Week 1: artifacts found, data verified, harness built
- 3 Oct: first API runs, B0 smoke test
- Nov 2026: email UTokyo professors
- Late Dec 2026: post to arXiv